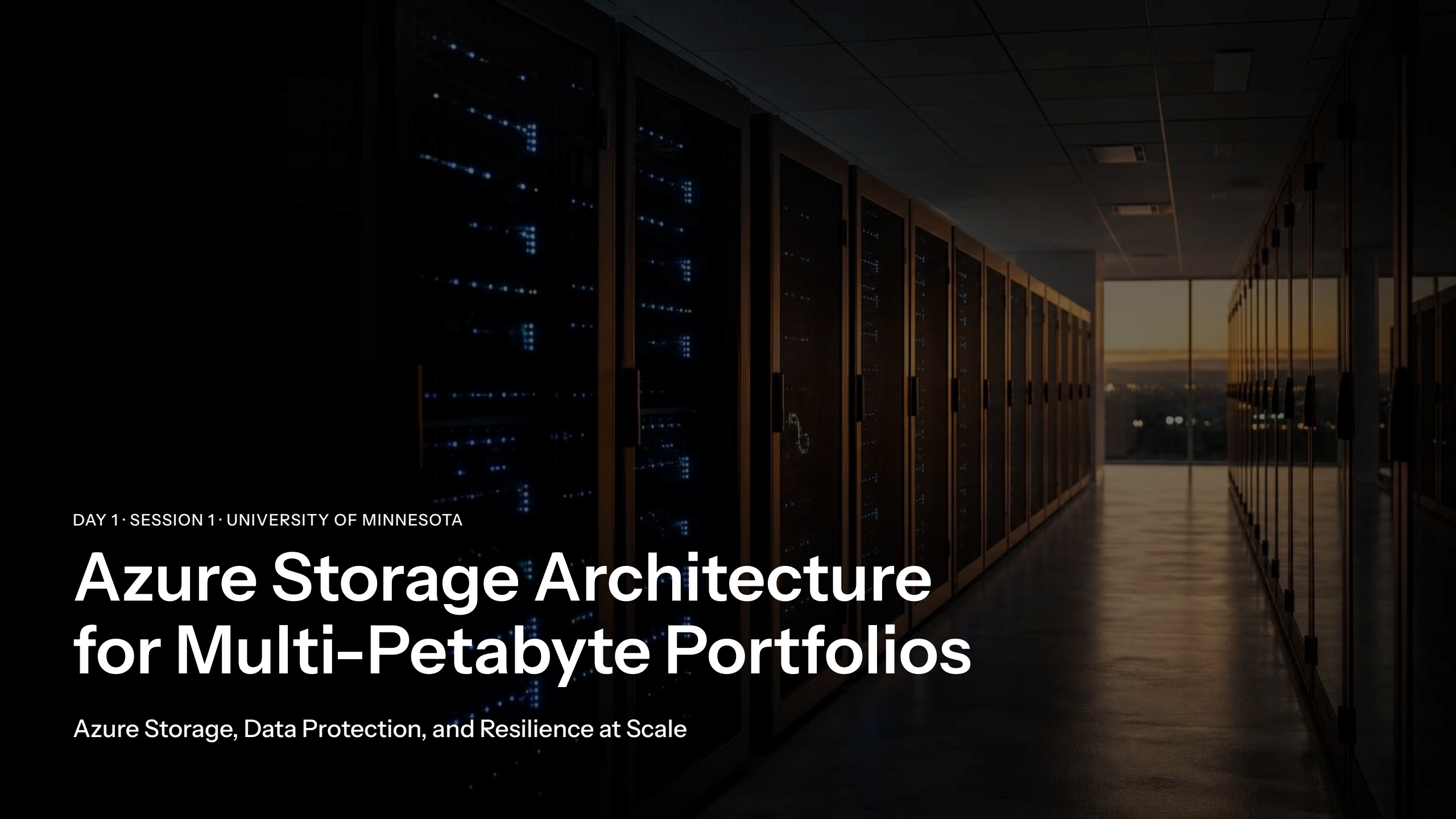


DAY 1 · SESSION 1 · UNIVERSITY OF MINNESOTA

Azure Storage Architecture for Multi-Petabyte Portfolios

Azure Storage, Data Protection, and Resilience at Scale

Today's four hours, block by block

01 **Block 1 · 60 min**

Storage account architecture and namespace decisions

02 **Block 2 · 50 min**

Access tiers, lifecycle management, and the tiering demonstration

Break · 15 min

03 **Block 3 · 50 min**

File, NAS, and HPC storage options; scale targets and sharding

04 **Block 4 · 65 min**

Design exercise and group presentations

Several petabytes, mostly research, some decisions you cannot undo

- The portfolio spans several petabytes, the majority research data
- It ranges from small humanities corpora to decades of polar satellite telemetry
- Goal: translate today's on-premises storage frames into Azure equivalents, know where scale limits sit, and spot the decisions that are hard to reverse once data lands

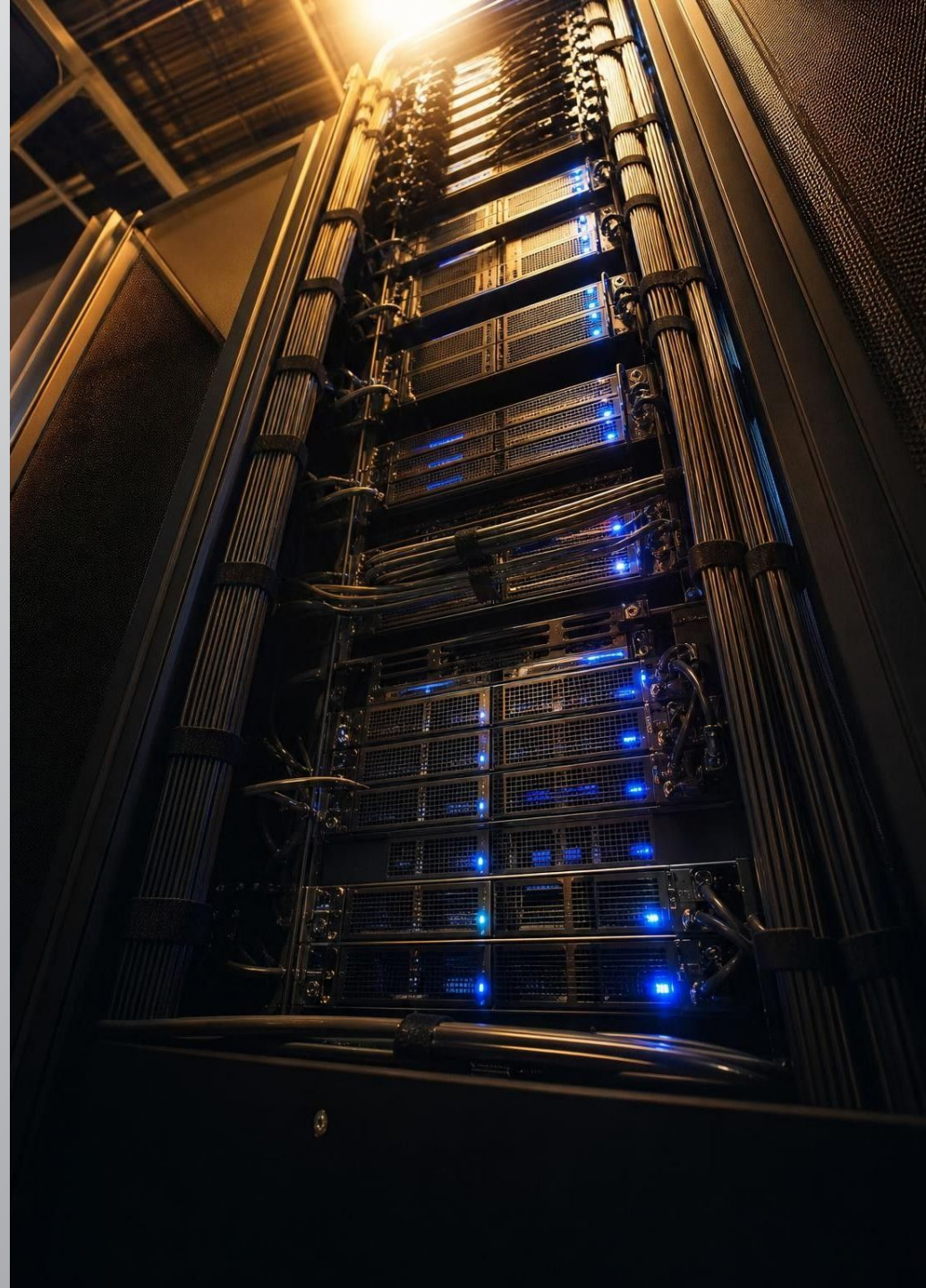
By the end of this session you can

- Select account type, performance tier, and namespace model for a workload
- Explain Hot, Cool, Cold, and Archive behavior, cost profile, and rehydration
- Design lifecycle policies that tier data by access pattern and age
- Position Azure Files, Azure NetApp Files, and Azure Managed Lustre against on-premises NAS and HPC scratch
- Identify scale targets and the patterns that work within them at petabyte scale

BLOCK 1 · 60 MIN

Storage account architecture and namespace decisions

Account type, redundancy, namespace, and naming: the choices that set the shape of everything after.



From storage frames to storage accounts

On-premises	Azure equivalent
Array or frame	Storage account: the unit of capacity targets, redundancy, networking, and keys
Volume or file system	Container or file share
LUN-level replication	Account redundancy and object replication
Array tiering	Blob access tiers and lifecycle policies

No capacity to buy up front: you pay for what is stored, transacted, and transferred.

Start every design from the account type

DEFAULT

Standard general-purpose v2

Recommended for blobs, files, queues, and tables; the only standard type that supports all four blob access tiers plus smart tier. GPv1 and legacy Blob storage accounts reached full retirement on October 13, 2026; anything left is being auto-migrated to GPv2, and GPv2 transaction pricing differs, so re-check cost models on migrated accounts.

PREMIUM

Premium block blob

SSD-backed, low latency and high transaction rates for small objects and analytics. No access tiers; LRS and ZRS only.

PREMIUM

Premium page blob

Page blobs only, for workloads that still depend on random-write VHD-style blobs. Most virtual machine disk scenarios now use managed disks.

PREMIUM

SSD (premium) file shares

FileStorage account kind. Required for NFS shares. Microsoft now describes Azure Files by media tier, SSD (premium) and HDD (standard), and offers a standalone file share resource (Microsoft.FileShares) for SSD NFS shares that needs no storage account at all.

PREVIEW OF SESSION 4

Redundancy is chosen per account

LRS

3 copies in one datacenter.

ZRS

3 copies across availability zones in one region.

GRS / GZRS

LRS or ZRS in the primary plus asynchronous copy to the paired secondary region.

RA-GRS / RA-GZRS

Adds read access to the secondary.

Premium accounts support LRS and ZRS only. Choose redundancy per data class, which drives account boundaries.

Decide carefully: some choices are hard or impossible to reverse

- Account kind and performance tier are set at creation; moving standard to premium means migrating data to a new account
- Enabling hierarchical namespace on an existing account is a one-way upgrade
- Redundancy is account-level, so mixed durability needs mean separate accounts
- Account names are permanent and appear in every endpoint URL

Treat these as architecture decisions, not configuration settings.

Hierarchical namespace: when a data lake is the right shape

Azure Data Lake Storage Gen2 adds real directories, atomic directory rename and delete, and POSIX-style ACLs on top of Blob Storage.

1 Right for

Research data lakes on Spark, Hadoop, Databricks, or Synapse, and directory-level permissions.

2 Still tiered

Supports access tiers and lifecycle management like flat namespace accounts.

3 Check first

Review the Blob feature support matrix; a few features have limitations on HNS.

Flat namespace or hierarchical: choosing per workload

CHOOSE FLAT

Object workloads

Backup targets, web content, and archives addressed by name. Every Blob data-protection feature is available: versioning, change feed, point-in-time restore, object replication, version-level WORM, and generally available blob index tags.

CHOOSE HNS

Analytics and directories

Analytics pipelines, directory renames at scale, per-folder ACLs for research groups. Azure RBAC is evaluated first, then POSIX ACLs on directories and files. NFS 3.0 and SFTP on Blob Storage require HNS.

⚠ **Know what you give up with HNS** Not supported on hierarchical namespace accounts: blob versioning, change feed, object replication, point-in-time restore, version-level WORM, page blobs. Preview only: blob index tags, blob snapshots. Still supported: container-level WORM, lifecycle management, soft delete, last access tracking, blob inventory, and Azure Backup vaulted backup for Data Lake Storage.

This decides whether a dataset's protection path is Blob-native (flat) or vault-based (HNS). Sessions 2, 3, and 4 build on this choice.

Naming and partitioning shape performance for a decade

Account names

3-24 lowercase letters and numbers, globally unique.

The name is the partition key

Blob Storage partitions by account, container, and blob name.

Spread hot writes

Avoid timestamp or sequential prefixes that concentrate traffic on one partition range; use a hash or varied prefix.

One convention

Account per environment and data domain, container per research unit or retention class.

BLOCK 1 CHECKPOINT

Three questions before data lands

01

Performance

Does this data need premium latency, or is standard GPv2 enough?

02

Namespace

Will anyone run analytics or need directory-level permissions on it?

03

Redundancy

What redundancy does its value justify, and does that force a separate account?

BLOCK 2 · 50 MIN

Access tiers, lifecycle management, and the tiering demonstration

Where the cost engine lives, and how to model it before a rule runs.

Four tiers, one trade: storage cost against access cost

1

Hot · online

Highest storage, lowest access cost; frequent access.

2

Cool · online

Infrequent access, minimum 30 days.

3

Cold · online

Rare access, fast retrieval, minimum 90 days.

4

Archive · offline

Latency of hours, minimum 180 days.

Capacity limits apply per account, not per tier; tiers apply to block blobs only.

Matching research data to tiers

Tier	Research data that belongs there
Hot	Active project data, current analysis inputs and outputs
Cool	Completed projects still consulted, recent instrument runs
Cold	Reference datasets read a few times a year but needed quickly
Archive	Retained-for-record data, decades of telemetry, preservation copies with hours-tolerant access

Minimum days

30	90	180
Cool	Cold	Archive

Minimum retention and early deletion change the math

- Deleting, overwriting, or moving a blob out early incurs a prorated early deletion charge
- Default account tier can be Hot, Cool, or Cold; Archive is set per blob
- Per-blob tier overrides the account default; blobs without an explicit tier inherit it
- Smart tier, now generally available, moves inactive blobs from Hot to Cool after 30 days and to Cold after 90, and promotes them back to Hot on access; no tier-change, retrieval, or early deletion charges apply inside smart tier
- Smart tier requires a GPv2 account with ZRS, GZRS, or RA-GZRS and supports block blobs only. It never uses Archive, ignores lifecycle tiering rules, bills access at Hot rates plus a monthly monitoring charge per 10,000 objects over 128 KiB, and is excluded from reserved capacity

Rehydration: plan for hours, not minutes

Archived blobs are offline. Rehydrate by Copy Blob to a new online blob, or by Set Blob Tier.

01

Standard priority

Up to 15 hours for objects under 10 GB.

02

High priority

May complete in under one hour for objects under 10 GB, at higher cost.

03

Signal completion

Event Grid notifies downstream automation when rehydration finishes.

Archive smart: package small files and know the edge cases

Bundle small files

Many small blobs rehydrate slower than fewer large ones due to per-blob overhead.

Snapshots and versions

Archived ones cannot be rehydrated in place; copy them to a new online blob.

Cross-account copies

Supported only within the same region.

Raise priority

Standard can be raised to High while a request is pending.

Lifecycle management: rules, filters, actions

Filters

Blob types, prefix match, and blob index tag match.

Conditions

Days since modification, creation, or last access.

Actions

Tier to Cool, Cold, or Archive, and delete, for base blobs, versions, and snapshots.

The policy is read and written in full, and Azure Storage runs it once per day.

Anatomy of a lifecycle rule

Example: move block blobs not modified in 30 days to Cool and in 90 days to Archive, scoped by prefix.

1

Rule

Name, enabled flag, type "Lifecycle".

2

Filters

Blob type and prefix scope.

3

baseBlob

Actions on the current version.

4

Versions, snapshots

Separate action blocks.

Author in the portal list view, code view, CLI, PowerShell, or Bicep.

Target tags, not folder paths

Blob index tags

Classify data by research unit and retention class independent of where it sits. Preview only on HNS accounts; use prefixes and directories there.

Last access tracking

Enables "not read in N days" rules; the Last accessed condition requires it. It cannot be enabled on an account with immutable storage, so archival accounts with WORM policies use creation or modification conditions instead.

Watch auto-tier to Hot

Data returning before 30 days can trigger early deletion fees; analyze access patterns first.

Model the money before you enable the rule

01 Inventory capacity, object counts, and last access by dataset (blob inventory reports help at scale)

02 Classify each dataset with tags

03 Price the steady-state capacity per tier

04 Add one-time tier-change transaction costs and any early deletion exposure

05 Review the affected datasets by name, then commit

LIVE DEMONSTRATION

Modeling a 3 PB research portfolio across tiers

01

Load

UMN Research Storage Inventory workbook (synthetic) into the cost model.

02

Tag

Datasets by research unit and retention class.

03

Apply

A tag-targeted lifecycle rule in a live subscription.

04

Compare

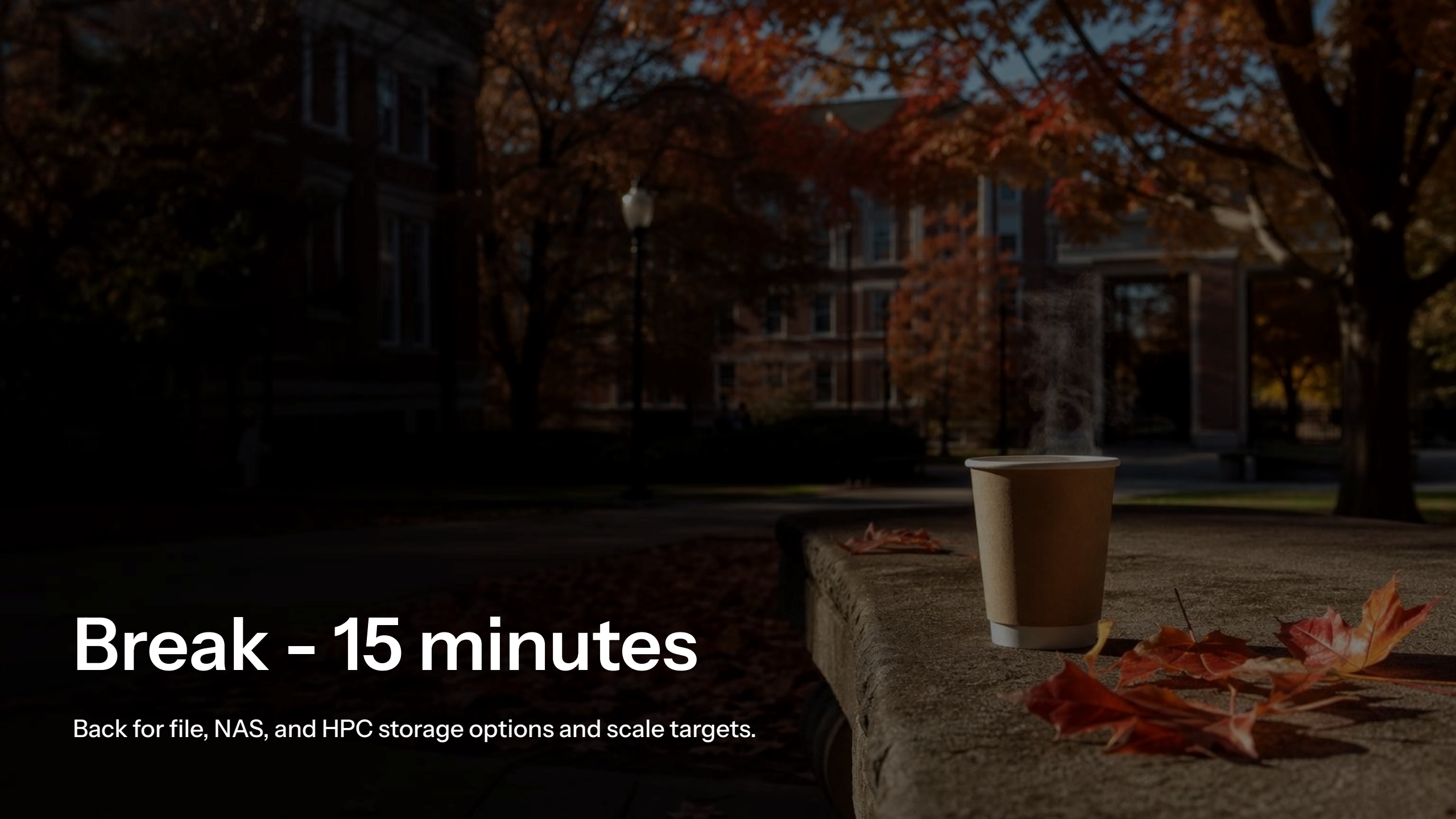
Projected monthly and annual cost, before and after.

Watch for the moment the projected cost drops and the affected datasets appear by name before the rule is committed.

What to look for in the demo

The cost model and the live rule run side by side. Track the projected figure as tags are applied.

Nothing commits until the affected datasets are listed by name and reviewed.



Break - 15 minutes

Back for file, NAS, and HPC storage options and scale targets.

BLOCK 3 · 50 MIN

File, NAS, and HPC storage options; scale targets and sharding

Matching file services to workloads, then designing within account limits.

Three file services, three different jobs

Azure Files

Fully managed SMB and NFS shares.

Azure NetApp Files

Enterprise NAS on NetApp ONTAP for latency-sensitive workloads.

Azure Managed Lustre

Managed parallel file system for HPC scratch.

Azure Files: managed shares for departments and hybrid sites

Media tiers and protocols

- SSD (premium) and HDD (standard); NFS shares require SSD
- Identity-based SMB access with AD DS or Microsoft Entra (Session 2)

Billing models: the FinOps decision

- Provisioned v2: storage, IOPS, and throughput set separately; recommended for all new shares; up to 256 TiB, with bursting
- Provisioned v1: legacy, SSD only; IOPS and throughput follow storage
- Pay-as-you-go: legacy, HDD only; transaction-based, hard to budget
- Reservations cover SSD v1 and HDD pay-as-you-go, not v2

Hybrid and protection

- Azure File Sync caches shares on on-premises Windows Servers with cloud tiering; Microsoft recommends HDD provisioned v2 shares for File Sync
- Share snapshots provide point-in-time read-only copies; vaulted backup for Azure Files is generally available (Session 3)

Azure NetApp Files: enterprise NAS performance

Service levels

Standard, Premium, Ultra, and Flexible, setting throughput relative to provisioned capacity.

Structure and protocols

Capacity pool, then volumes; NFS v3 and v4.1, SMB, and dual-protocol.

Protection built in

Snapshots, cross-region and cross-zone replication, and backup.

Strong fit

Databases (including Oracle, Session 5), SAP, and engineering applications.

Azure Managed Lustre: HPC scratch with a Blob backbone

- Fully managed Lustre parallel file system for high-throughput compute clusters
- Links to a Blob container to import data before a job and export results after
- Treat it as scratch: Blob holds the durable copy, Lustre provides the speed
- Maps to today's HPC scratch tiers supporting research computing

Mapping what you run today to Azure

On-premises today	Azure fit	Why
Departmental SMB file shares	Azure Files (SMB)	Managed shares, AD-based access, File Sync caching
Linux NFS exports for research groups	Azure Files NFS or Azure NetApp Files	NFS semantics; NetApp Files for higher performance
Tier-1 NAS for databases and apps	Azure NetApp Files	Low latency, selectable service levels
HPC parallel scratch	Azure Managed Lustre	Parallel throughput, Blob integration
Bulk research and archive data	Blob Storage / ADLS Gen2	Tiering to Cool, Cold, Archive at scale

Scale targets: the numbers behind the sharding decision

Target (standard GPv2, default)	Value
Storage accounts per region per subscription	250 by default, 500 by request
Maximum account capacity	5 PiB by default, higher by request
Request rate per account	40,000 requests per second in major regions, including Central US, North Central US, East US, East US 2, West US 2, and West US 3; 20,000 elsewhere
Ingress per account	60 Gbps in those regions; 25 Gbps elsewhere
Egress per account	200 Gbps in those regions; 50 Gbps elsewhere
Private endpoints per account	200
Single block blob	Up to about 190.7 TiB; target 3,000 requests per second per blob

High-end targets, not guarantees; results depend on object size and access pattern, so test your own. At a partition's limit you see 503 Server Busy or 500 Operation Timeout: retry with exponential backoff and avoid sequential names that create hot partitions.

Sharding patterns for portfolios beyond one account

By data domain

Shard by research unit so cost, access, and redundancy map cleanly.

By protection class

Shard by redundancy, since redundancy is account-wide.

By request profile

Separate high-throughput analytics from archive accounts to isolate request rates.

Catalog and template

Map datasets to accounts, and deploy every account from the same template.

Security baseline for every account

Encrypt

Always on at rest (256-bit AES); Microsoft-managed keys or customer-managed keys in Key Vault or Managed HSM.

Transport

Require secure transfer and a minimum TLS version of 1.2.

Azure Storage stopped accepting TLS 1.0 and 1.1 in early 2026; set 1.2 explicitly in Bicep and Azure Policy so templates stay self-describing.

Identity

Disable anonymous blob access and, where possible, shared key access in favor of Microsoft Entra ID.

Network isolation: private endpoints and firewall rules

- Private endpoints give the account a private IP in your virtual network, reachable over ExpressRoute or VPN
- Set the firewall default action to deny, then allow selected networks
- Disable public network access where all clients are private
- Plan private DNS zones so endpoint names resolve to private IPs

BLOCK 4 · 65 MIN

Design exercise and group presentations

Apply the session's decisions to three real university data classes.

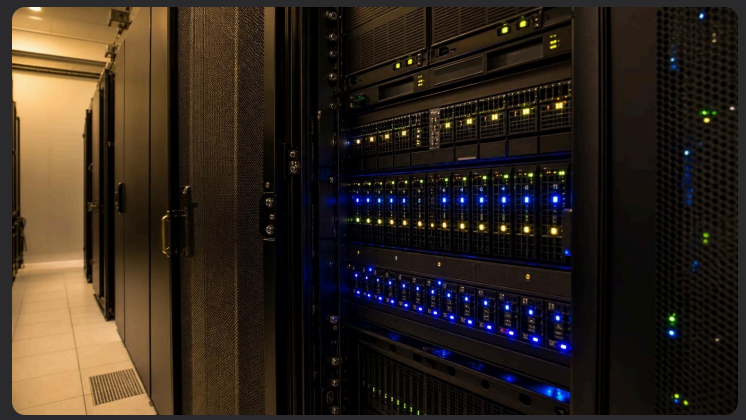
Design exercise: mapping three university data classes

Polar satellite archives

Decades of telemetry, rarely read, irreplaceable.

Humanities text corpus

Active researchers, moderate size.



Enterprise application data

Business-critical, frequent access.

For each class specify account type, namespace, tier strategy, redundancy, lifecycle rules, and a cost line.

How we will critique: Well-Architected Framework lenses

Reliability

Redundancy matched to the data's value

Security

Baseline controls applied by default

Cost optimization

Tiering and lifecycle with a stated rationale

Operational excellence

Repeatable, template-driven deployment

Performance efficiency

Account and service matched to access pattern

Key takeaways and what you take into Session 2

Architecture and cost

Account type, namespace, and redundancy are architecture decisions; tiers and lifecycle are the cost engine.

Materials

Storage decision matrix reference card, design worksheet, tiering demo script, Research Storage Inventory workbook.

Next: Session 2

Long-term archival, data movement at scale, and hybrid identity.

Capstone input

Tagged, lifecycle-managed research storage with reserved capacity

Well-Architected Framework lenses

Reliability: Redundancy matched to the data's value.

Security: Baseline controls applied by default.

Cost optimization: Tiering and lifecycle with a stated rationale.

Operational excellence: Repeatable, template-driven deployment.

Performance efficiency: Account and service matched to access pattern.